
Decoding the DNA of Real Madrid

A Football Analytics Investigation into La Liga Playing Styles, Cluster Structure, and Match Outcome Prediction

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2025/26 Season — SP1.csv

Tools

Python·scikit-learn·pandas·matplotlib·seaborn

“Football is not just about winning — it’s about understanding what separates the great from the rest.”

Contents

1 Introduction	3
2 Dataset and Preparation	3
2.1 Overview	3
2.2 Why the Betting Columns Were Dropped	3
2.3 Building Team DNA	4
3 Understanding Team Identity	4
4 Discovering Playing Styles	4
4.1 Feature Selection	4
4.2 K-Means and K-Means++	4
4.3 Hierarchical Clustering	5
4.4 DBSCAN — Is Anyone an Outlier?	5
5 The Real Madrid Case Study	5
5.1 Cluster Membership	5
5.2 Nearest and Furthest Neighbours	5
5.3 What Makes Real Madrid Distinct	6
6 Understanding Success in La Liga	6
6.1 Model Performance	6
6.2 What Drives Match Outcomes	7
6.3 Model-by-Model Notes	7
7 Final Insights	7
7.1 Answering the Question	7
7.2 A Reflection	8

1. Introduction

Football has always been a game of opinions. Managers disagree, pundits argue, fans debate endlessly. But the game produces data — shots, corners, goals, fouls — and that data tells a story opinions often miss.

This project started from a personal question: *what actually makes Real Madrid different from other clubs in La Liga, and which teams are statistically most similar to it?* As a Real Madrid supporter who wants to work in football analytics, I used the tools from this course to investigate something I genuinely wanted to know.

La Liga makes for a compelling case study because it is so unequal. The gap between a top club and a relegation side is enormous, but the interesting territory lies in between — which teams cluster together, which play similarly, and which factors actually separate the top from the rest.

Central question: “What makes Real Madrid statistically different from other La Liga clubs, and which teams are most similar to it?”

Each algorithm was used as a tool to answer a specific football question — not applied mechanically because it appeared on the syllabus.

2. Dataset and Preparation

2.1. Overview

The dataset covers the full La Liga 2025/26 season: 380 matches across 20 clubs, with 131 columns per match. The vast majority of those columns are betting odds, not football statistics. **Table 1.** Dataset at a Glance

Property	Value
Matches	380 (complete season)
Clubs	20
Football-stat columns	23
Betting/odds columns	108 (dropped)
Missing values (football stats)	0
Result split	H48.9% · D24.5% · A26.6%

2.2. Why the Betting Columns Were Dropped

108 of 131 columns are bookmaker odds. They were excluded for three reasons:

- **Data leakage:** odds are set after team news, encoding information unavailable at prediction time.
- **Redundancy:** all bookmakers price similarly — ten columns saying the same thing.
- **Wrong focus:** the goal is to understand playing styles, not predict markets.

2.3. Building Team DNA

The raw data has one row per match. Clustering needs one row per club. All statistics were pooled across home and away appearances into per-game averages, creating a 20-row *team DNA matrix* covering attack, defence, discipline, and efficiency.

3. Understanding Team Identity

Each club’s profile was built from four groups of metrics:

- **Attacking:** goals scored/game, shots on target/game, conversion rate
- **Defensive:** goals conceded/game, SOT allowed, clean sheet rate
- **Discipline:** fouls/game, yellow cards/game, red cards/game
- **Efficiency:** shot accuracy, win rate (points per game was dropped — $r > 0.97$ with win rate)

Table 2 shows Real Madrid against the league average.

Table 2. Real Madrid vs. La Liga Average — Key Metrics				
Metric	Real Madrid	League Avg	Diff	Rank
Goals scored / game	2.026	1.312	+0.714	2/20
Goals conceded / game	0.921	1.370	0.449	1/20
Shots on target / game	6.658	4.213	+2.445	2/20
Win rate	0.711	0.360	+0.351	2/20
Clean sheet rate	0.368	0.230	+0.138	2/20

Real Madrid ranks first or second on every key metric — not through extreme advantage in any single area, but through consistent excellence across all of them. No other club achieves that across the board.

Figure 1 illustrates this using a normalised radar chart.

Figure 1. Real Madrid’s normalised performance radar vs. the La Liga average. The gold area exceeds the league average on all seven dimensions.

4. Discovering Playing Styles

4.1. Feature Selection

Before clustering, a correlation audit removed redundant metrics: goal difference (linear combo of GF and GA), half-time goals (overlap with full-time), and points per game (near-duplicate of win rate). The final feature set has **11 variables**, all standardised to zero mean and unit variance.

4.2. K-Means and K-Means++

Both the elbow method and silhouette score pointed to $K = 4$, matching a natural football taxonomy: Elite, Contenders, Mid-table, Survival. K-Means++ improved over standard K-Means:

Table 3. K-Means vs. K-Means++ Performance ($K = 4$)		
Method	Inertia	Silhouette

K-Means (random init)	85.45	0.1647
K-Means++ (smart init)	82.59	0.2144

Table 4. K-Means++ Cluster Assignments

Cluster	Teams
Elite	Atlético de Madrid · Barcelona · Real Madrid
Contenders	Betis · Celta · Villarreal
Mid-table	Getafe · Vallecano
Survival	Alavés · Ath. Bilbao · Elche · Espanyol · Girona · Levante · Mallorca · Osasuna · Oviedo · Sevilla · Sociedad · Valencia

The Elite cluster has only three members. That reflects the actual structure of La Liga — the gap between these clubs and the rest is not just points, it is statistical.

Figure 2 shows all 20 clubs projected onto the two principal components that explain 69.3% of variance. The Elite cluster forms a clearly separated island; Real Madrid is at its edge. Figure 2. All 20 clubs in PCA space, coloured by K-Means++ cluster. Real Madrid () sits at the boundary of the Elite group, slightly more isolated than Barcelona and Atlético.

4.3. Hierarchical Clustering

Ward linkage clustering was run without pre-specifying *K*, as a blind check on the K-Means solution. The dendrogram (Figure 3) recovers the same four groups. Real Madrid and Barcelona merge first at the shortest Ward distance; Atlético joins slightly later, reflecting its more defensive profile. Figure 3. Ward linkage dendrogram. The red dashed cut at distance 6 produces four clusters identical to K-Means++. Real Madrid (gold label) merges with Barcelona before any other pairing.

4.4. DBSCAN — Is Anyone an Outlier?

DBSCAN identifies clubs that do not belong to any dense region. With $\epsilon = 3.16$ and `min_samples=2`, it found **one outlier: Getafe**. Real Madrid was assigned to a cluster — it is elite, but not statistically anomalous. Getafe’s physical, foul-heavy, low-scoring approach places it in a statistical corner of its own. This adds useful nuance: Real Madrid is extraordinary, but its profile is a coherent extreme of a dimension that also includes Barcelona and Atlético. Getafe’s style has no peer.

5. The Real Madrid Case Study

5.1. Cluster Membership

Every clustering method places Real Madrid in the same Elite group. That consistency — across K-Means++, hierarchical clustering, and DBSCAN — is not a coincidence. It is a structural feature of the data: Real Madrid exists in a different statistical region from seventeen other La Liga clubs.

5.2. Nearest and Furthest Neighbours

Euclidean distances in the 11-dimensional standardised feature space identify Real Madrid’s closest statistical siblings.

Table 5. Real Madrid — Nearest Neighbours by Statistical Distance

Rank	Club	Distance	Profile match
1	Barcelona	2.237	High scoring, high SOT, solid defence
2	Atlético de Madrid	2.831	Similar win rate, more defensive style
3	Betis	4.891	Moderate — different defensive profile
4	Villarreal	4.954	Moderate — similar attacking output
19	Sevilla (<i>furthest</i>)	8.027	Low scoring, low win rate, bottom zone

The gap between neighbours is revealing. Real Madrid to Barcelona: 2.24 units. Real Madrid to Sevilla: 8.03 units — more than three times wider. The inequality of La Liga is visible directly in these distances.

5.3. What Makes Real Madrid Distinct

The answer is not one exceptional statistic. It is the combination:

- **Best defensive record in the league:** 0.921 goals conceded/game (rank 1/20), better even than Atlético
- **Top-2 attacking output:** 2.026 goals scored/game, 6.658 SOT/game
- **Win rate nearly double the league average:** 0.711 vs. 0.360
- **Clean sheets at 60% above average:** 0.368 vs. 0.230

No other club appears near the top on all four simultaneously. Atlético has the defensive record but not the attack. Barcelona scores heavily but concedes more. The combination is what separates Real Madrid.

6. Understanding Success in La Liga

The classification task asks a different question: *given match statistics, which factors most strongly predict the final result?* Six models were trained on an 80/20 split of the 380 matches using 22 features (shots, corners, fouls, cards, half-time statistics, and derived differences).

6.1. Model Performance

Figure 4. All six models compared by accuracy and F1 Weighted on the 76-match test set. The dotted line marks the naive baseline of always predicting a home win (48.9%).

Table 6. Model Performance Summary — Test Set (76 matches)

Model	Accuracy	F1 Macro	F1 Weighted
Logistic Regression	63.2%	0.6037	0.6467
AdaBoost	60.5%	0.5285	0.5804
Random Forest	59.2%	0.5247	0.5841

KNN ($K = 19$)	59.2%	0.5167	0.5710
Stacking	53.9%	0.4566	0.5237
Decision Tree	44.7%	0.4203	0.4599

Logistic Regression leads at 63.2%. A linear model outperforming 300 trees is unexpected — but it makes sense here: once you know the half-time goal difference and the shots-on-target balance, the relationship with the final result is largely linear. Adding complexity introduces variance without reducing bias.

6.2. What Drives Match Outcomes

Figure 5. Random Forest feature importances (300 trees). Half-time statistics dominate — confirming that half-time leads in La Liga are almost always maintained.

Both the Decision Tree and Random Forest converge on the same top predictors:

1. **Half-time goal difference** — strongest single predictor
2. **SOT difference** (home minus away shots on target)
3. **Half-time result** (encoded)
4. Home shots on target (absolute)
5. Away fouls committed

The football interpretation: teams that lead at half-time almost always win, and the side generating more quality chances nearly always comes out on top. None of this is surprising to a football fan — but the models *quantify* it. For Real Madrid, which leads at half-time in 71% of its matches and ranks second in shots on target, this is a structural advantage built into every game it plays.

6.3. Model-by-Model Notes

KNN ($K = 19$): proximity-based, reasonable at 59.2% but limited by the small dataset. **Decision**

Tree (depth 3): most interpretable, struggles with draws. **Logistic Regression**: regularised

($C = 0.5$), balanced classes — best overall. **Random Forest** (300 trees): confirms feature importances are stable, not overfitting. **AdaBoost** (200 rounds): converges in 50 rounds, solid at 60.5%. **Stacking**: KNN+DT+LR → RF meta-learner underperforms with only 76 test instances — not enough for the meta-learner to learn when to trust which base model.

7. Final Insights

7.1. Answering the Question

After applying ten machine learning algorithms to a full La Liga season, the answer to the central question is clear — and it is not a single number.

Real Madrid is statistically distinct because it leads in *every* relevant dimension simultaneously. Best defensive record (1st/20), top-2 attack, top-2 shots on target, top-2 win rate, top-2 clean sheets. No other club appears at the top of all four dimensions at once.

Both K-Means++ and hierarchical clustering place Real Madrid in an Elite cluster of three.

DBSCAN confirms it is not an outlier — just the high end of a coherent dimension shared with Barcelona and Atlético. The nearest-neighbour analysis identifies Barcelona as the closest statistical twin (distance

2.24), while the furthest club — Sevilla — sits 8.03 units away, more than three times the Real Madrid–Barcelona gap.

The classification models consistently identify the same success factors: half-time lead and shot-on-target advantage. Real Madrid structurally dominates both.

7.2. A Reflection

Machine learning does not reveal things about football that a careful analyst would never work out — but it quantifies them in ways that qualitative analysis cannot. The claim that Real Madrid and Sevilla are “very different” is imprecise. The distance of 8.03 units in a standardised 11-dimensional space is specific.

The most valuable lesson from this project is about data decisions, not algorithms. Choosing to drop 108 betting columns, to pool home and away statistics into a single fingerprint, to use shot-on-target ratio rather than raw shot count — these choices shaped the results far more than the choice between K-Means and hierarchical clustering. The algorithm is usually the easy part. Understanding what the data is actually measuring is where the real work is.

All analyses: Python3, scikit-learn, pandas, matplotlib, seaborn. Dataset: La Liga 2025/26, SP1.csv (Football-Data.co.uk format).

Hala Madrid.